Historical years, territory and the estimation sample

Controls: Historical territory, territory year and instrument year.

1. The two historical years have different roles. The territory year determines historical sample availability and historical fixed-effect groups. In State analysis it also determines the regions over which counties are aggregated. The instrument year chooses the historical population used in the regression. Both offer census years from 1790 to 1900. GDP and employment come from the separately selected modern year.

Territory year 1790 with instrument year 1900 uses the 1790 grouping and requires usable historical inputs in both years. Changing the instrument year does not change regional boundaries. OLS retains the historical availability restriction used for the IV comparison.

2. State observations follow the groups in the dataset. The refreshed County historical fixed-effect assignments define the regions. *Historical States Only* uses the dataset's states-only assignments. *Historical States & Territories* uses its full assignments. A region is one group of counties with the same historical identity. Counties without an assignment are excluded.

For 1790 the dataset contains 13 states-only groups and 16 full-scope groups: the same 13, Northwest Territory, Southwest Territory and Vermont. Vermont remains a separate group in the full file; it is absent from the states-only file. These are the dataset's definitions, rather than a count of today's states.

3. Aggregate the same counties throughout. Within each group, sum modern county GDP, employment and land area, and construct the CH index from those counties. Thus the 1790 Massachusetts group includes Maine counties, while Virginia includes Kentucky and West Virginia counties. The assignment uses county centres, with the dataset's interior-point fallback; whole counties are assigned, not split at historical state borders. Combined employment-data units remain together.

Standard CH, county adjustments, NLS and every IV/GMM construction use this same regional membership. County and MSA analysis retain their own observation units and use these identities for historical grouping and sample selection.

4. The eligible groups need not all enter a regression. Every State specification uses the same historical coverage choice. In each selected historical year, divide the area with a recorded population by the full mapped area of the region's member counties. Recorded zeros count toward known area. The default is *No coverage filter*. Select 80% or 95% to require that share in both years. County matching and area weighting sum the usable population records. Historical CH needs positive population and a finite index. A prediction also needs the inputs for the selected density and weights. Missing modern GDP, employment, land area or a selected county adjustment can reduce the sample further.